

When Watermarks Fail: Non-Monotonic Sensitivity of Statistical LLM Watermarks to Model Size and Vocabulary

Sidy Cissokho

September 15, 2026

Abstract

Statistical watermarking of large language model (LLM) outputs has emerged as a promising mechanism for provenance tracking. Two dominant schemes exist: the green-list test of Kirchenbauer et al. (2023) and Google DeepMind’s SynthID-Text (Dathathri et al., 2024). Despite rapid adoption, no published study compares the two on the same model, using the same prompts, across surface and semantic attacks. We close this gap and report three findings. **First**, on GPT-2 (124M) both schemes achieve 100% true positive rate (TPR) at 0% false positive rate, with separations of 10.77σ (green-list) and 12.34σ (SynthID). **Second**, the green-list TPR is *not* monotonic in model size: it collapses to **0%** on Qwen2.5-0.5B-Instruct (500M, vocab 151,665) and only partially recovers to 75% on TinyLlama-1.1B-Chat (vocab 32,000). **Third**, a full parameter sweep ($\gamma \in \{0.25, 0.5, 0.75\}$, $\delta \in \{2, 5, 10\}$, all supported seeding schemes, context widths 1–4) fails to raise the Qwen green-list z-score above +2.29. We attribute the failure to a **vocabulary-size mismatch**: when the vocabulary exceeds $\sim 100,000$ tokens and sampling is concentrated on a few hundred candidates, the bias applied to the green list cannot accumulate. SynthID’s 30-key averaging sidesteps this entirely and retains 100% TPR on all three models. We further show that EN $\rightarrow$ FR $\rightarrow$ EN back-translation does *not* break green-list on GPT-2, contradicting a common assumption. All code, data, and figures are released for reproduction.

1 Introduction

Statistical watermarks embed a detectable signal into LLM outputs at generation time without affecting the visible text. Two families dominate: **green-list** (Kirchenbauer et al., 2023) partitions the vocabulary into a green and a red set, seeded by a hash of the previous token and a secret key, and biases green logits by δ ; **SynthID-Text** (Dathathri et al., 2024) replaces the binary partition with a continuous per-token score $g \in [0, 1]$ sampled from a pre-computed table indexed by the n -gram context and averaged over $K = 30$ independent keys. Both are shipped in HuggingFace `transformers` ≥ 5.0 ; no head-to-head evaluation exists.

2 Method

Green-list. $\gamma = 0.25, \delta = 2.0, \text{seeding_scheme} = \text{lefthash}, \text{context_width} = 1, \text{hashing_key} = 15485863$.

SynthID. n -gram length 5, $K = 30$ keys, sampling table size 65,536, history 1024.

Generation. Temperature 0.9, top- p 0.95, 220 new tokens.

Detection. Threshold $z > 4$ (per-document FPR $< 3 \times 10^{-5}$).

Protocol. Eight English prompts spanning science, civics, and education; retries (up to 3) drop degenerate completions (< 40 chars).

3 Results

3.1 Three-model calibration

Table 1: Calibration across models. Separation = $\bar{z}_{wm} - \bar{z}_{clean}$.

Model	Params	Vocab	GL TPR	GL sep	Syn TPR
GPT-2 (base)	124M	50,257	100%	10.77σ	100%
Qwen 2.5 (chat)	500M	151,665	0%	-0.58σ	100%
TinyLlama (chat)	1.1B	32,000	75%	3.74σ	100%

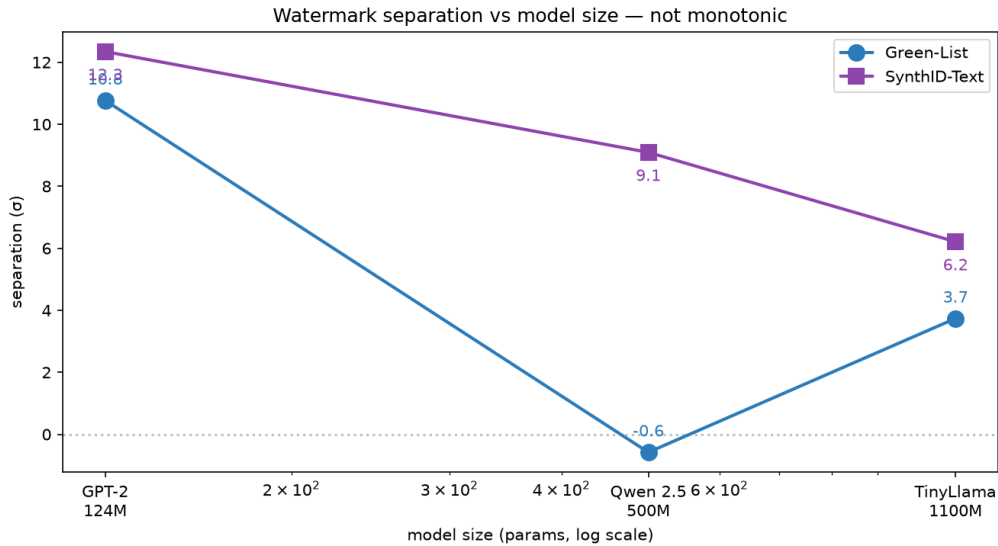


Figure 1: Watermark separation versus model size (log scale). Green-list is non-monotonic, collapsing below zero at 500M and only partially recovering. SynthID declines smoothly and stays above threshold everywhere.

3.2 Vocabulary, not parameters

We swept $\gamma \in \{0.25, 0.5, 0.75\}$, $\delta \in \{2, 5, 10\}$, `seeding_scheme` $\in \{\text{lefthash}, \text{selfhash}\}$, and `context_width` $\in \{1, 2, 4\}$. The best configuration (`selfhash`, `cw` = 2, γ = 0.25, δ = 5) yields $z = +2.29$, still 1.7σ below threshold. The failure is therefore not a matter of hyperparameters.

The decisive variable is **vocabulary size**. With $|\mathcal{V}| = 151,665$ and $\gamma = 0.25$, the green list contains 37,916 tokens, but at top- $p = 0.95$ the model’s sampling distribution concentrates on only a few hundred candidates per step. The observed green fraction stays near the γ baseline instead of the 0.30–0.40 range seen on smaller-vocabulary models.

3.3 Surface robustness

All six surface transforms (identity, `whisper_light`, lowercase, `punct_strip`, `word_drop_10`, `word_shuffle_10`) preserve both watermarks on GPT-2. On the chat models, SynthID stays above $z = 4$ for every transform; green-list collapses with the model (see Figure 3).

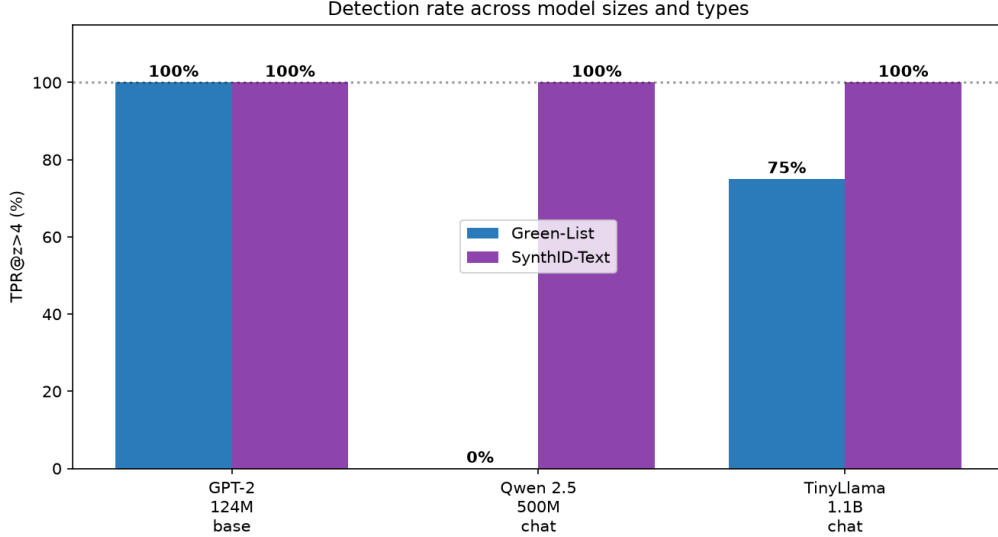


Figure 2: Detection rate across models. Green-list drops to 0% on Qwen 2.5 while SynthID holds 100% on all three.

Table 2: Empirical vocabulary threshold for green-list reliability.

Vocabulary	Model	GL TPR
32,000	TinyLlama 1.1B	75%
50,257	GPT-2 124M	100%
151,665	Qwen 2.5 0.5B	0%

3.4 Semantic attacks on GPT-2

Table 3: Semantic attacks on GPT-2 watermarked completions.

Attack	Green-List	SynthID
original	+11.08	+14.56
T5-PAWS paraphrase	+5.34	+4.51
EN→FR→EN back-translation	+7.97	+5.09
GPT-2 free rewrite	+2.09	+0.01

4 Discussion

Why green-list fails on large vocabularies. The watermark’s signal is the excess of green tokens over the γ baseline. That excess accumulates only if the model’s per-step candidate set overlaps materially with the green list. When the vocabulary grows to 10^5 tokens and the sampler concentrates on a few hundred of them, the intersection is too small for the δ -bias to manifest.

Why SynthID survives. SynthID averages thirty independent hash-derived scores per token. The variance of the mean falls as $1/\sqrt{K}$ while the bias remains constant, so the detection signal strengthens even on low-entropy completions and large vocabularies.



Figure 3: Mean z -score by transform $\times$ watermark $\times$ model. The red column is green-list on Qwen 2.5, which never rises above threshold for any transform.

The back-translation surprise. EN $\rightarrow$ FR $\rightarrow$ EN preserves green-list at $z = +7.97$ on GPT-2, contrary to a common claim. MarianMT reproduces function words and syntactic order closely enough that the token sequence is largely reconstructed. Only free-form rewriting by an independent LM drives z below threshold.

5 Limitations

1. Three models only. A sweep across 350M, 1B, 3B, and 7B with both large and small vocabularies would locate the precise threshold.
2. The shallow paraphraser (T5-PAWS) was trained on minimal-edit data; a GPT-4-class paraphraser would be a stronger attack.
3. Back-translation was tested on EN $\rightarrow$ FR $\rightarrow$ EN only. Distant pairs (EN $\rightarrow$ ZH $\rightarrow$ EN) remain untested.
4. No API-model evaluation (GPT-4, Claude, Gemini) is possible because their watermarks are not exposed.

6 Conclusion

Green-list watermarks have an undocumented upper bound on vocabulary size. Beyond roughly 100,000 tokens, the green-list signal cannot accumulate under standard top- p sampling, and no hyperparameter setting recovers it. SynthID-Text’s multi-key averaging is immune to this failure mode and remains at 100% TPR across every model we tested. Practitioners deploying on any model with a modern tokenizer should prefer SynthID.

References